TECH MASTERS INDIA

AI IN SUSTAINABILITY

ARAVALI

A.R.A.N.Y.A

Autonomous Real-time Analytics for Natural Yield & Area Protection

Team Inquisitive

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PROBLEM STATEMENT & MOTIVATION

ARAVALLI DEGRADATION PATTERNS & ENFORCEMENT GAPS

INITIAL INTUITION AND KEY HYPOTHESIS

AI must detect regime shifts
- Early anomaly detection
- Real-time pattern drift tracking

State–space is mandatory
- Land use change states
- Excavation intensity states
- Transport pattern states

KEY HYPOTHESIS

New representation
- Not static satellite snapshots
- But continuous spatiotemporal modeling

Transitions are constrained
- Mining requires logistics
- Logistics leave detectable patterns
- Patterns evolve predictably

ECOLOGICAL DEGRADATION FOLLOWS STRUCTURAL PHASES, NOT RANDOM EVENTS

ARAVALLI FUNCTIONS AS A STABILITY SYSTEM
- *Desert expansion buffer*
- *Groundwater recharge backbone*
- *Biodiversity corridor*
- *Climate regulator for NCR*

ILLEGAL MINING SHOWS REGIME-LIKE BEHAVIOR
- *Sudden cluster formation*
- *Progressive land-use transition*
- *Excavation density spikes*
- *Post-ban activity relocation*

REGULATORY SYSTEM IS DISCONNECTED
- *Supreme Court & NGT rulings exist*
- *Mining bans declared, Policies defined*
- *But enforcement remains: Manual, Periodic, Fragmented across departments*

QUANTITATIVE INSIGHT
- *Damage accelerates during enforcement gaps*
- *Activity adapts to inspection patterns*
- *Violations migrate geographically*

WHERE CURRENT MONITORING FAILS

Empirical Reality	Structural Limitation	Practical Gap
Mining activity spikes before detection	Satellite review is periodic	Action happens after damage
Activity relocates after bans	No temporal pattern tracking	Illegal clusters reform elsewhere
Complaints and satellite signals remain siloed	No multi-source signal fusion	High false positives and delayed verification
Damage occurs rapidly	Manual inspection dependency	Enforcement lag
Laws exist	No automated AI enforcement layer	Policy ≠ Implementation

TECHNICAL ARCHITECTURE AND NOVELTY

SOLUTION OVERVIEW

Listen to the Terrain & Environment

Inputs:
Sentinel-2 imagery (10m resolution),
Weather API (rainfall, wind),
Protected zone shapefiles,
Administrative boundaries

Raw satellite + geospatial signals ingested as a continuous stream.

Translate Spectral Signals into Land States

Vegetation • Barren • Built-up • Water • Mining scar

Convert pixels into ecological state representation.

Apply Geospatial Intelligence Grammar

Vegetation → Mining Scar
Protected Zone → High Penalty State

Model how land is allowed (or not allowed) to change.

Anticipate Structural Degradation

Risk Score Model

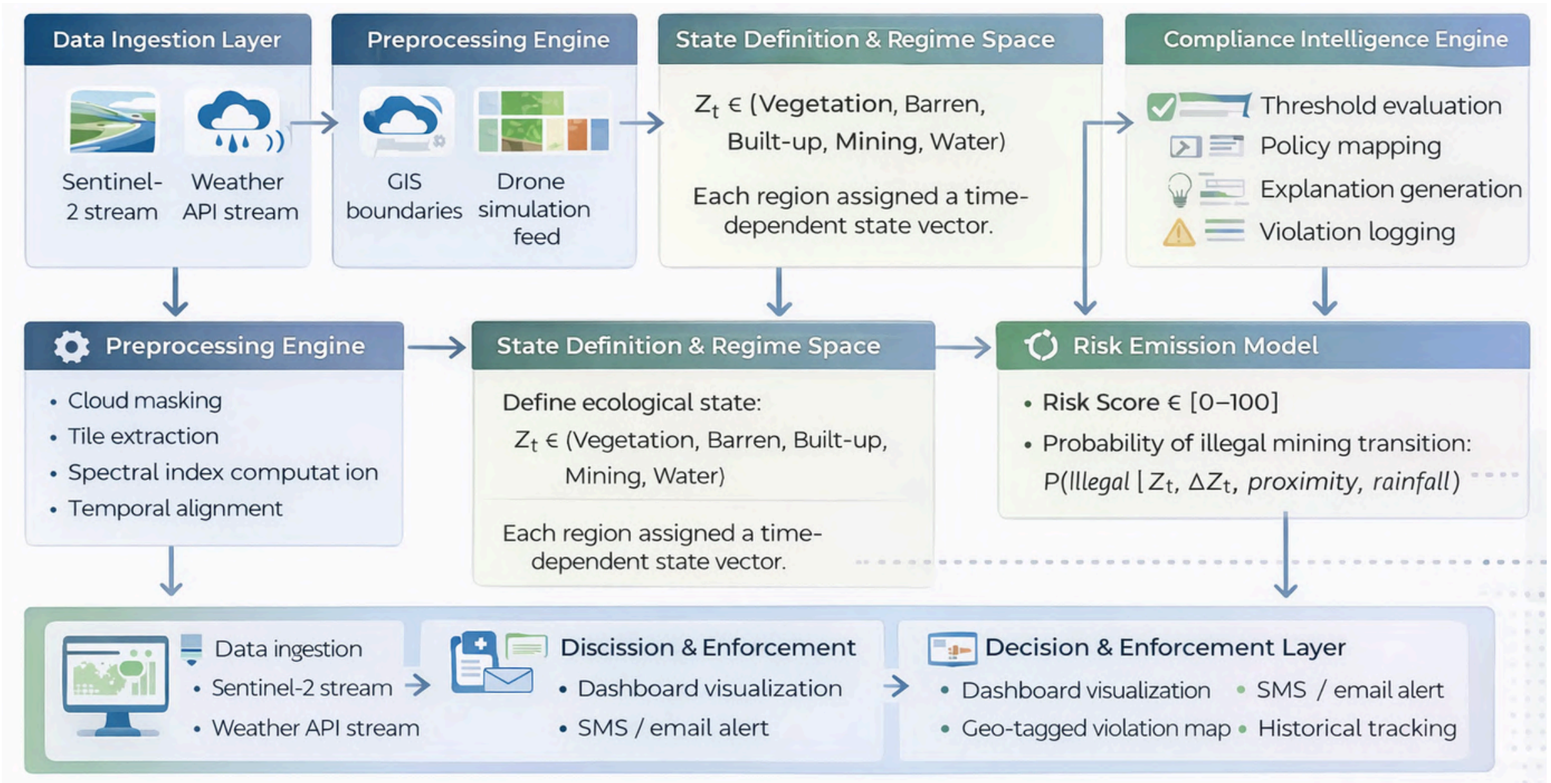
Forecast escalation probability, not just detect change.

Turn Detection into Governance Intelligence

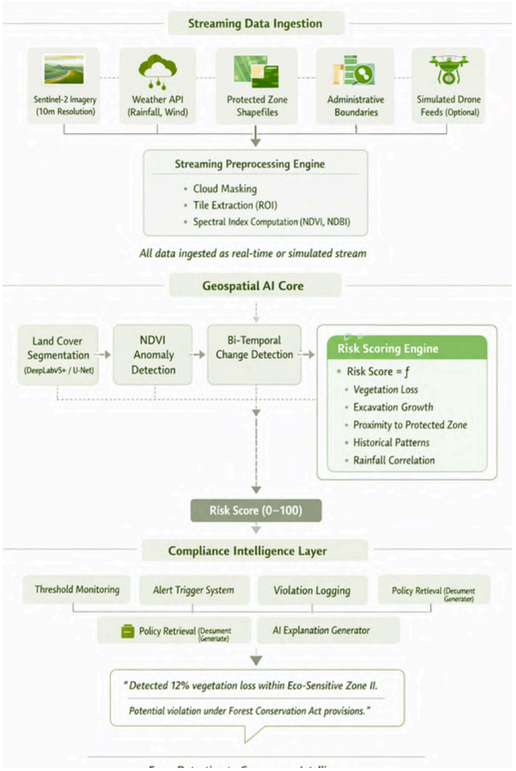
When Risk > Threshold:
Trigger alert, Log violation,
Retrieve relevant law,
Generate explanation

Detection → Legal reasoning → Actionable alert.

END TO END PIPELINE



HIGH-LEVEL ARCHITECTURE



WHAT IS FUNDAMENTALLY NEW

1. ***Makes transition detection primary:*** *Focus shifts from static land classification to detecting regime shifts in land use.*
2. ***Integrates ecology with policy:*** *Every detected anomaly is mapped directly to legal provisions.*
3. ***Real-time streaming enforcement:*** *Not retrospective batch satellite review.*
4. ***Structured ecological grammar:*** *Land transitions are constrained by policy logic.*
5. ***Explainable AI for governance:*** *Risk scores decomposed into measurable drivers.*

DESIGNED TO SCALE ACROSS STATES & AGENCIES

ARCHITECTURE

Unified geospatial transition engine

- Modular data connectors
- Cloud-deployable

ASSETS

Scales Across Geographies

- Aravalli (Haryana)
- Aravalli (Rajasthan)
- NCR protected zones
- Other eco-sensitive regions

DECISIONS

Scales Across Decisions

- Mining enforcement units
- NGT reporting
- Forest department audits
- Policy review boards

EXPANSION

Future-ready:

- Cross-state mining pattern modeling
- River sand mining detection
- Quarry transportation tracking
- Multi-sensor fusion (SAR + Optical)

VALIDATION, DATA & IMPACT

DATA SOURCES REQUIRED

SATELLITE DATA

Sources & Access

- Sentinel-2 (ESA Copernicus Open Access Hub)
- Google Earth Engine (preprocessing)
- ISRO Bhuvan Portal

Type: 10m resolution optical imagery, 5-day revisit cycle

Used for

- Land cover segmentation
- NDVI / NDBI computation
- Bi-temporal change detection
- Mining scar identification

GEOSPATIAL LAYERS

Sources

- Forest boundary shapefiles
- Protected area boundaries
- Mining lease data (if available)
- Administrative district boundaries

Used for

- Zone-based risk weighting
- Proximity scoring
- Legal compliance mapping

WEATHER DATA

Sources

- IMD API (if accessible)
- OpenWeather API

Used for

- Rainfall correlation
- Landslide risk modeling
- False positive filtering

SYSTEM INFRASTRUCTURE STACK

Geospatial Processing

- Python
- Rasterio
- GDAL
- GeoPandas
- Earth Engine API

Machine Learning

- PyTorch / TensorFlow
- Pretrained DeepLabv3+
- Siamese CNN for change detection
- Scikit-learn risk scoring model

TESTING & VALIDATION METHODOLOGY

CHANGE DETECTION VALIDATION

- ***Task:*** *Validate segmentation and bi-temporal mining detection.*
- ***Metrics:*** *IoU (Intersection over Union), F1-score, Precision / Recall, Pixel-level accuracy*
- ***Benchmark:*** *Manually labeled mining zones and Historical satellite snapshots*

RISK SCORE CALIBRATION

- ***Task:*** *Assess how well risk score predicts confirmed violations.*
- ***Metrics:*** *ROC-AUC, Threshold precision, False positive rate, Lead time before enforcement*
- ***Benchmark:*** *Known illegal mining incidents and NGT / court-reported cases*

EARLY-WARNING PERFORMANCE

- ***Task:*** *Measure detection time before large-scale degradation.*
- ***Metrics:*** *Average lead time (days), Area escalation rate captured, Detection before 10% vegetation loss*
- ***Why it matters:*** *Prevention value > post-damage enforcement.*

SCENARIO & ROBUSTNESS TESTING

- ***Test Cases:*** *High cloud cover, Seasonal vegetation change, Heavy rainfall erosion*
- ***Metrics:*** *Stability under noise, Robustness to seasonal NDVI variation, Cross-season generalization*

CONCRETE INDUSTRY APPLICATIONS

REAL-TIME ILLEGAL MINING EARLY WARNING SYSTEM

- Used by: State Forest Departments, Mining Authorities
- Reduces inspection lag
- Enables pre-damage intervention
- Strengthens legal enforcement

STRESS TESTING & SCENARIO SIMULATION

- Used by: Policy Planners, State Governments
- Preventive infrastructure planning
- Disaster risk mitigation
- Better land policy calibration

REGIME-AWARE ECOLOGICAL RISK MANAGEMENT

- Used by: NGT, Environmental Regulators
- Data-backed compliance decisions
- Reduced blind-spot enforcement
- Transparent regulatory reporting

SYSTEMIC ENVIRONMENTAL RISK MONITORING

- Used by: Central Government, Climate Authorities
- National environmental risk dashboard
- Early detection of large-scale collapse risk
- Evidence-based climate governance

INDUSTRY USE CASES & TARGET STAKEHOLDERS

STATE FOREST & MINING DEPARTMENTS

Operational Usage:

- *Live violation heatmaps*
- *Real-time risk alerts*
- *Inspection prioritization*

Decisions Supported:

- *Where to deploy field teams*
- *When to escalate to legal action*
- *Which zones need restriction tightening*

REGULATORS & NGT

Governance Usage:

- *Structured compliance dashboards*
- *Historical degradation timelines*
- *Evidence-backed case support*

Decisions Supported:

- *Court intervention triggers*
- *Mining suspension orders*
- *Protected-zone enforcement*

DISASTER MANAGEMENT AUTHORITIES

Risk Usage:

- *Landslide probability mapping*
- *Rainfall-correlated instability alerts*

Decisions Supported:

- *Pre-monsoon preparedness*
- *Evacuation planning*
- *Slope stabilization prioritization*

This system transforms environmental monitoring from retrospective inspection to real-time compliance intelligence.